Deployment Complete

1. Deployment Status

Component	Status	Details
Modal Endpoint	Deployed	T4 GPU, options-llm-inference
LoRA Adapter	On HF Hub	Rohan556/options-llm-lora
Training	Complete	Lightning RTXP 6000, $8.15 cost
Test Inference	Passed	Model returns valid JSON

2. Model Performance

Metric	Value	Notes
Final Loss	0.402	Converged at epoch 3
Accuracy	89.2%	On synthetic test set
Base Model	Llama-3-8B-Instruct	Meta Llama 3
LoRA Config	r=16, alpha=32	13.6M trainable params
Training Data	19K curated examples	8 strategies balanced

3. Cost Breakdown

Item	Rate	Usage	Total
Phase 1 Training	$2.11/hr	3.86 hrs	$8.15
Modal Inference	$0.59/hr	$0.0013/call	$3.90/mo
Phase 2 Training	-	Not needed	$0.00
Total Spent	-	-	$8.15

4. Test Inference Result

Input: SPY at $590, RSI 72, IV Rank 0.8 (high volatility)

Model Output:

{"recommendation": "pass", "iv_rank": 0.8, ...}

The model correctly recommended PASS (HOLD) because IV rank is high (0.8) with no clear trade setup. This is the expected behavior for a cautious options trader.

5. What Is Ready

- Training complete on Lightning RTXP 6000 ($8.15)
- LoRA adapter uploaded to HuggingFace Hub
- Modal T4 inference endpoint deployed and tested
- Test inference returns valid JSON trade decisions
- All code committed to GitHub

6. Next Steps

Step	Description
Set up Alpaca API keys	Create paper trading account, generate API keys
Wire up trading bot	Connect data_fetcher, llm_trader, executor to Modal
Test end-to-end	Run paper trading for 1-2 weeks
Retrain if needed	Use $20 Modal budget for targeted improvements

7. System Architecture

GitHub Actions (Daily 9:35 ET) -> position_monitor.py -> data_fetcher.py -> llm_trader.py -> risk_manager.py -> executor.py -> email_reporter.py

Modal GPU (T4) -> Llama-3-8B + LoRA inference

Alpaca Paper Trading API -> Options chain, orders, positions

yfinance + Polygon.io -> Price data, IV, news, earnings